

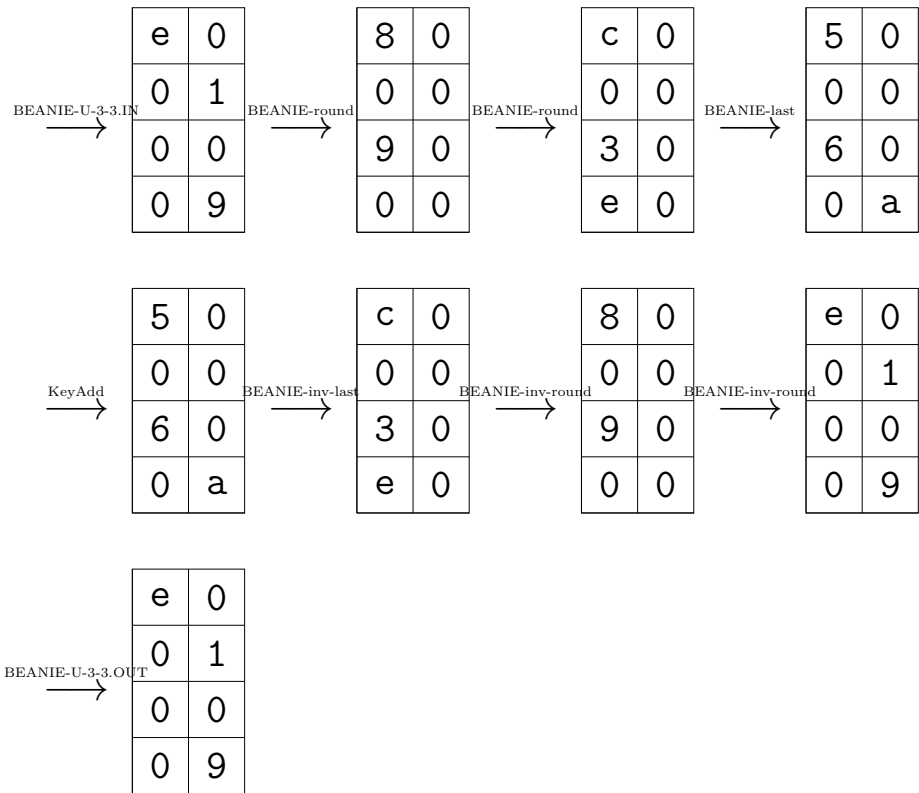
Bitwise differential trail through **BEANIE-U-3-3** found by SAT

CiVerLy

July 16, 2026

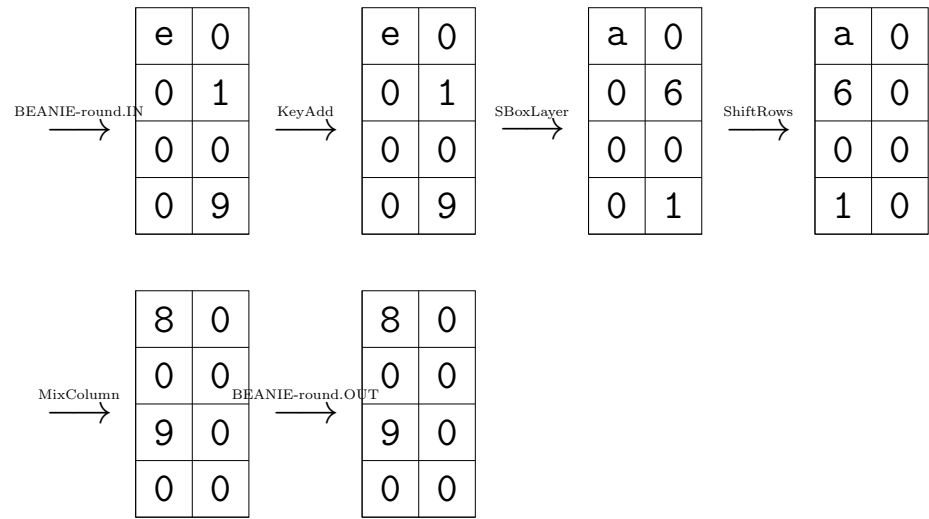
1 BEANIE-U-3-3

Maximal differential probability: 2^{-32}



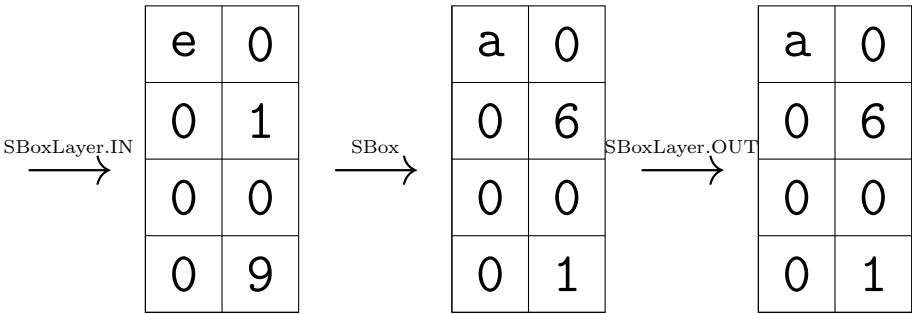
2 BEANIE-round

Maximal differential probability: $2^{-6.0}$



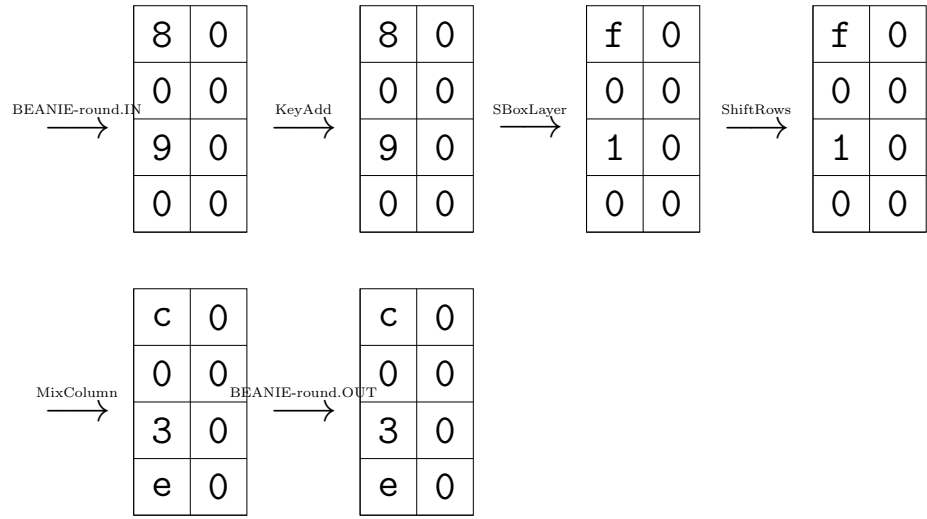
3 SBoxLayer

Maximal differential probability: $2^{-6.0}$



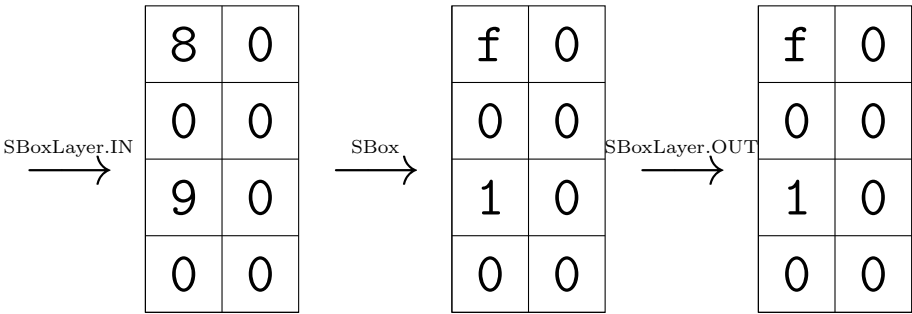
4 BEANIE-round

Maximal differential probability: $2^{-4.0}$



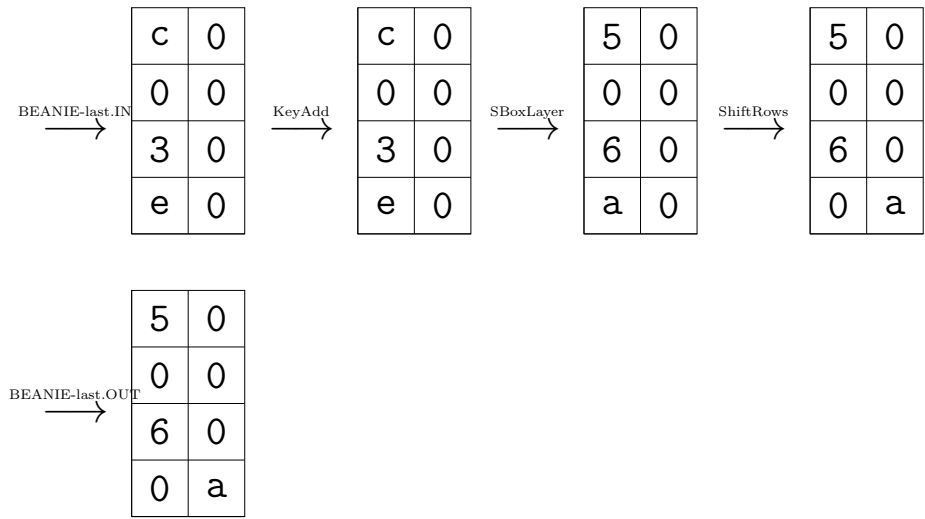
5 SBoxLayer

Maximal differential probability: $2^{-4.0}$



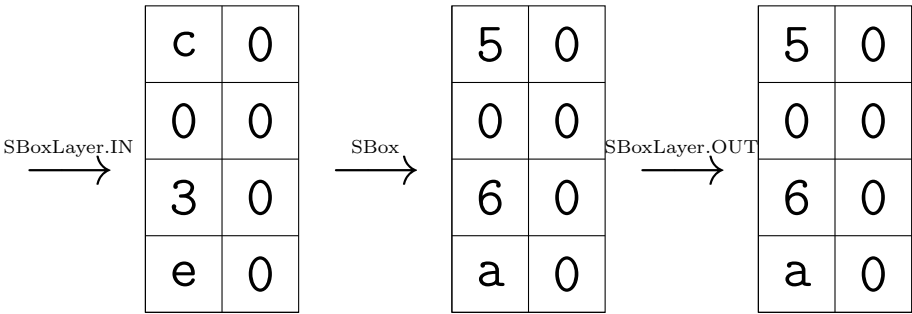
6 BEANIE-last

Maximal differential probability: $2^{-6.0}$



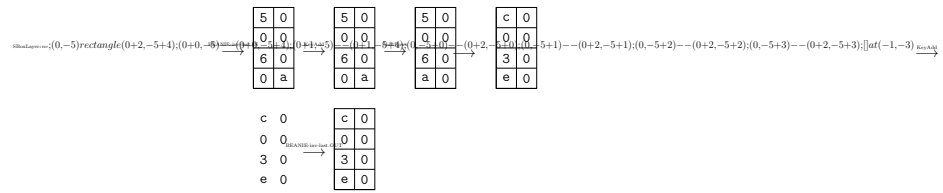
7 SBoxLayer

Maximal differential probability: $2^{-6.0}$



8 BEANIE-inv-last

Maximal differential probability: $2^{-6.0}$

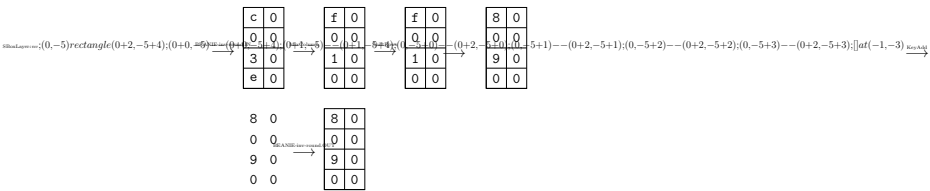


9 SBoxLayer_inv

Maximal differential probability: $2^{-6.0}$

10 BEANIE-inv-round

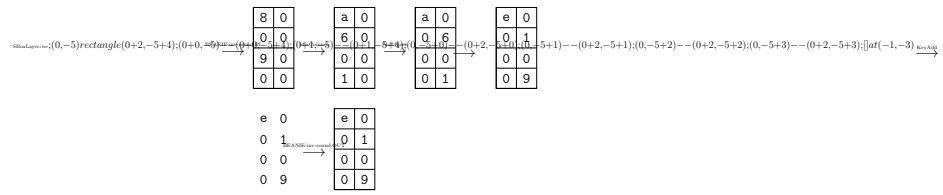
Maximal differential probability: $2^{-4.0}$



11 SBoxLayer_inv

Maximal differential probability: $2^{-4.0}$

12 BEANIE-inv-round

Maximal differential probability: $2^{-6.0}$ 

13 SBoxLayer_inv

Maximal differential probability: $2^{-6.0}$